Prescriptive Analytics in R

Course Project

Instructions:

This document is the second part of the two-part Course Project. Part One was completed in RStudio within the online version of the course. In this document, you will not be working in RStudio. Rather, you will be using the skills you have gained in Module Two of the course to determine an optimization approach to a problem.

Except as indicated, use this document to record all your project work and responses to any questions. At a minimum you will need to turn in a digital copy of this document to your instructor as part of your project completion. You may also have additional supporting documents that you will need to submit. Your instructor will provide feedback to help you work through your findings.

Note: Though your work will only be seen by those grading the course and will not be used or shared outside the course, you should take care to obscure any information you feel might be of a sensitive or confidential nature.

A submit button can be found on the Course Project assignment page. Information about the grading rubric is available on any of the course project assignment pages online. Do not hesitate to contact your instructor if you have any questions about the project.

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PART TWO

Determine and Formulate an Optimization Approach

You are a logistics manager for a large company that makes dog food. You are tasked with figuring out the best way to ensure sufficient supply at your eight distribution centers from three of your manufacturing facilities.

As their logistics manager, you need to figure out how to supply each of the three distribution centers with enough dog food to meet their demand in the best way possible. The three tables below show the data you'll need to make this decision. Note that not every manufacturing plant can reliably send dog food to every distribution center, and that these cases are marked as NA in Table Three.

Table One illustrates the demand at each of your eight distribution centers:

TABLE ONE

Distribution Center	Demand
Seattle	33,913
Atlanta	202,677
Denver	74,160
Louisville	45,108
Madison	99,146
Salt Lake	123,802
New Orleans	74,497
Pittsburgh	56,947

In Table Two, we show the capacity and manufacturing costs at three available plants:

TABLE TWO

	Buffalo	Tampa	Fort Worth
Capacity	206,704	255,446	325,524
Manufacturing Cost	$300	$334	$329

Table Three displays shipping costs per unit between each distribution center - plant pair. Note if the value is missing that indicates no reliable shipper if available for that route:

TABLE THREE

	Buffalo	Tampa	FortWorth
Seattle	$ 115.08	$ 127.23	$ 118.71
Atlanta	$ 91.75	$ 27.79	$ 62.64
Denver	$ 108.94	NA	$ 94.53
Louisville	$ 66.56	$ 59.14	$ 64.14
Madison	$ 70.66	$ 83.29	$ 72.56
SaltLake	$ 101.29	$ 119.77	$ 130.72
NewOrleans	NA	$ 53.08	NA
Pittsburgh	$ 31.79	$ 79.65	$ 78.79

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What objective are you, the logistics manager, trying to achieve in this situation?

It seems that we are minimizing the total cost of producing and shipping dog food from the manufacturing plants to the distribution centers. The total cost needs to be minimized while meeting other factors such as: the demand at each center not exceeding the capacity of production at each plant, and using valid shipping routes.

What is your objective function? Please give this answer in the form in which you'd input it into R.

```
#costs = manufacturing + shipping

costs <- c(415.08, 461.23, 447.71, 391.75, 361.79, 391.64, 408.94, 423.53, 366.56, 393.14, 393.14, 370.66, 417.29,
           401.56, 401.29, 453.77, 459.72, 387.08, 331.79, 413.65, 407.79)
```

How many constraints do you have, and what are they?

There are 11 constraints in total. 8 demand constraints and 3 supply constraints.

Demand Constraints:

Buffalo to Seattle + Tampa to Seattle + Fort Worth to Seattle = 33,913

Buffalo to Atlanta + Tampa to Atlanta + Fort Worth to Atlanta = 202,677

Buffalo to Denver + Fort Worth to Denver = 74,160

Buffalo to Louisville + Tampa to Louisville + Fort Worth to Louisville = 45,108

Buffalo to Madison + Tampa to Madison + Fort Worth to Madison = 99,146

Buffalo to Salt Lake + Tampa to Salt Lake + Fort Worth to Salt Lake = 123,802

Tampa to New Orleans = 74,497

Buffalo to Pittsburgh + Tampa to Pittsburgh + Fort Worth to Pittsburgh = 56,947

Supply Constraints:

Buffalo to Seattle + Buffalo to Atlanta + Buffalo to Denver + Buffalo to Louisville + Buffalo to Madison + Buffalo to Salt Lake + Buffalo to Pittsburgh <= 206,804

Tampa to Seattle + Tampa to Atlanta + Tampa to Louisville + Tampa to Madison + Tampa to Salt Lake + Tampa to New Orleans + Tampa to Pittsburgh <= 255,446

Fort Worth to Seattle + Fort Worth to Atlanta + Fort Worth to Denver + Fort Worth to Louisville + Fort Worth to Madison + Fort Worth to Salt Lake + Fort Worth to Pittsburgh <= 325,524

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```
dir<-c("==", "==", "==", "==", "==", "==", "==", "==", "<=", "<=", "<=")

rhs <-c(33913, 202677, 74160, 45108, 99146, 123802, 74497, 56947, 206704, 255446, 325524)
```

Submit the R code that defines your objective function, constraint matrix, dir vector, and rhs vector.

```
install.packages("ROI")
install.packages(c("Rglpk", "ROI.plugin.glpk))
library(ROI)
library(Rglpk)
library(ROI.plugin.glpk)

#Objective function
costs <- c(415.08, 461.23, 447.71, 391.75, 361.79, 391.64, 408.94, 423.53, 366.56, 393.14, 393.14, 370.66, 417.29,
           401.56, 401.29, 453.77, 459.72, 387.08, 331.79, 413.65, 407.79)

#Constraint matrix
L<-matrix(c(1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
            0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
            0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
            0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0,
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0,
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0,
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1,
            1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0,
            0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 1,
            0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1), nrow= 11, byrow = TRUE)

#dir vector and rhs vector
dir<-c("==", "==", "==", "==", "==", "==", "==", "==", "<=", "<=", "<=")
```


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```
rhs <-c(33913, 202677, 74160, 45108, 99146, 123802, 74497, 56947, 206704, 255446, 325524)

#Formulate linear program
LP<-OP( c(415.08, 461.23, 447.71, 391.75, 361.79, 391.64, 408.94, 423.53, 366.56, 393.14, 393.14, 370.66, 417.29,
          401.56, 401.29, 453.77, 459.72, 387.08, 331.79, 413.65, 407.79),
        L_constraint(L = matrix(c(1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
                                  0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
                                  0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
                                  0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
                                  0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0,
                                  0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0,
                                  0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0,
                                  0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1,
                                  1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0,
                                   0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 1,
                                   0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1), nrow= 11, byrow = TRUE),
                        dir = c("==", "==", "==", "==", "==", "==", "==", "==", "<=", "<=", "<="),
                        rhs =c(33913, 202677, 74160, 45108, 99146, 123802, 74497,
                               56947, 206704, 255446, 325524)),
                        max= FALSE)
LP
sol <- ROI_solve(LP,solver="glpk")
sol
sol$solution
```

To submit this assignment, please refer to the instructions in the course.

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